

CIFAR-10 CNN Report: ReLU vs. Tanh

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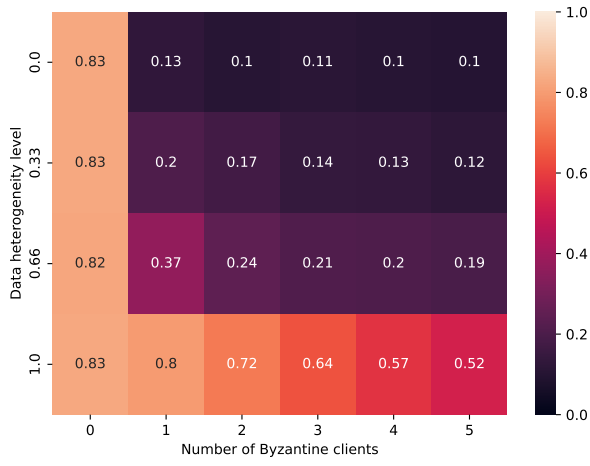
1 Introduction

This report compares two CIFAR-10 CNN configurations side by side:

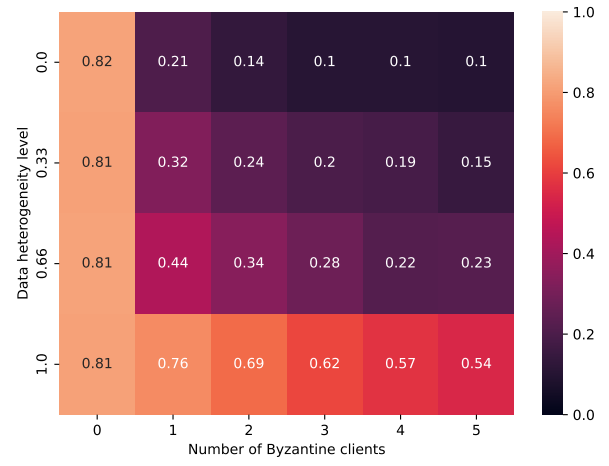
1. **CNN ReLU**: Standard CNN with ReLU activations, no clipping.
2. **CNN Tanh**: CNN with Tanh activation (implicit saturation).

Both share the same DSGD training setup, NNM+ARC pre-aggregation, and $f = 0.5$ / $\gamma = 0.1$ sweep grid.

2 Best Overall Performance (Worst-Case Across Attacks)



(a) CNN ReLU

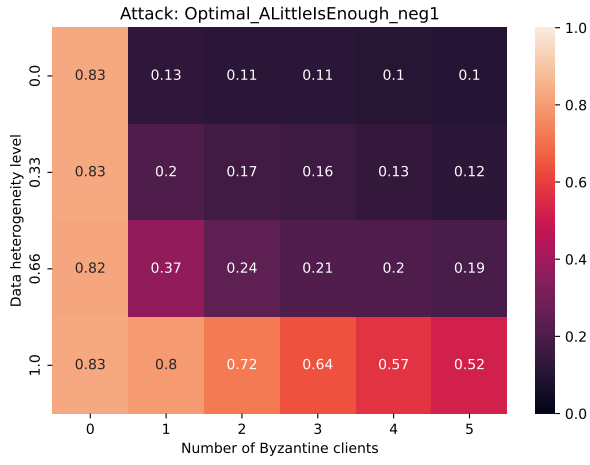


(b) CNN Tanh

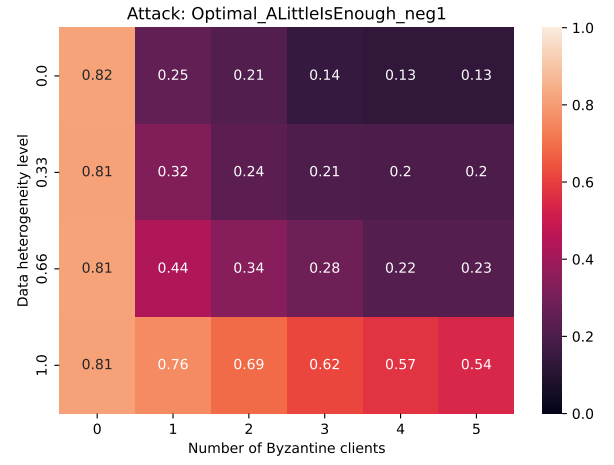
Figure 1: Best overall test accuracy (worst-case across attacks).

3 Best Aggregator under Specific Attacks

3.1 Optimal ALIE Attack



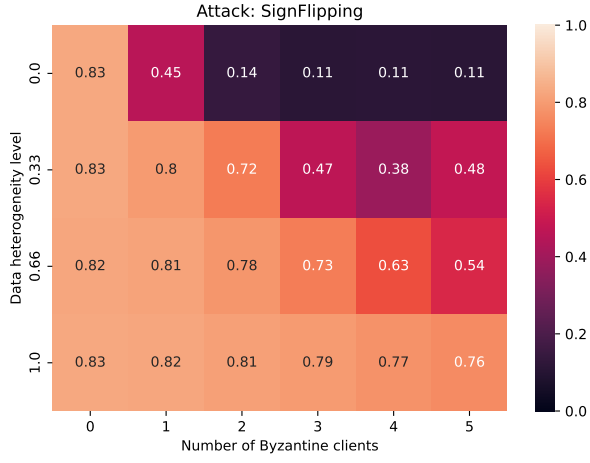
(a) CNN ReLU



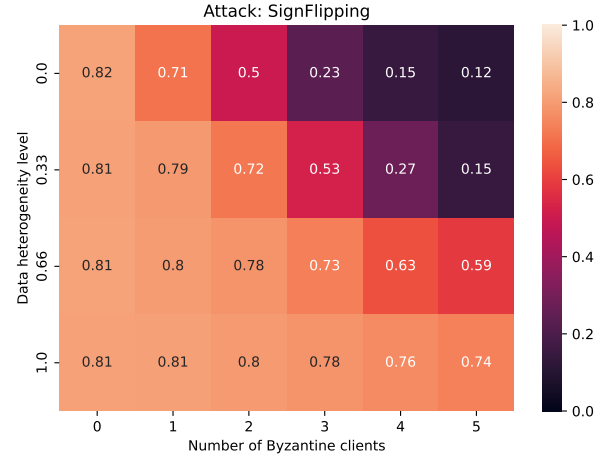
(b) CNN Tanh

Figure 2: Best test accuracy under ALIE.

3.2 Sign Flipping Attack



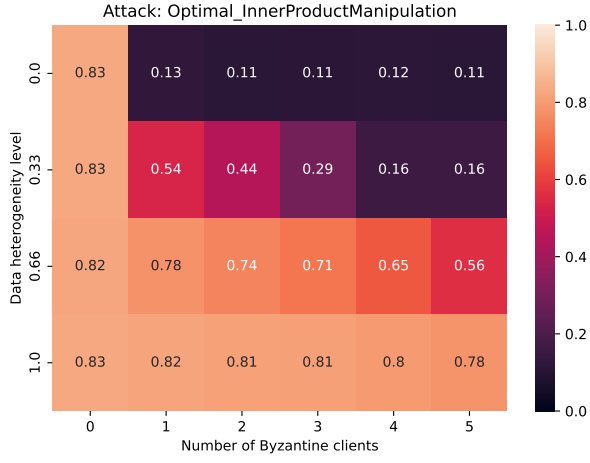
(a) CNN ReLU



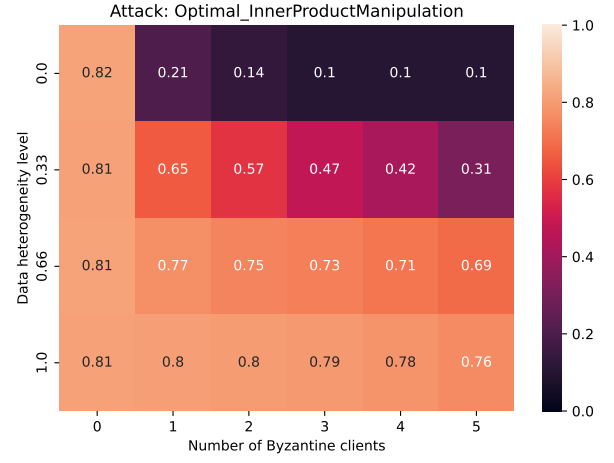
(b) CNN Tanh

Figure 3: Best test accuracy under SF.

3.3 Optimal IPM Attack



(a) CNN ReLU

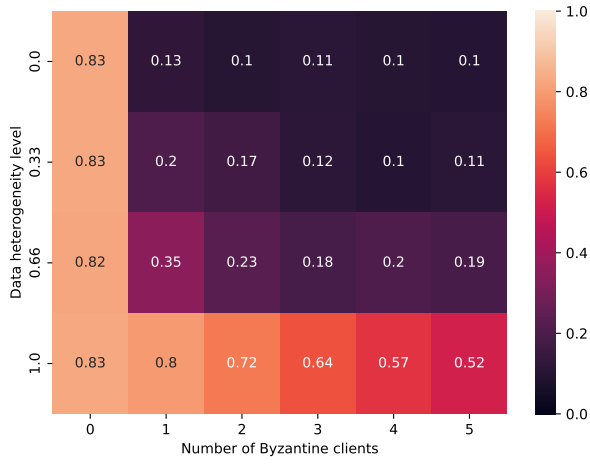


(b) CNN Tanh

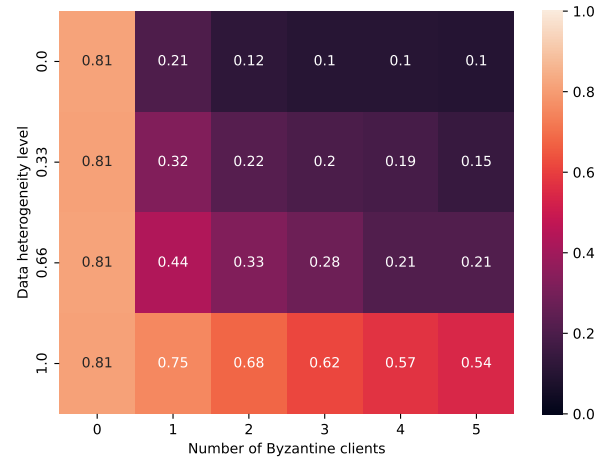
Figure 4: Best test accuracy under IPM.

3.4 Centered Clipping (CC)

3.4.1 CC under Worst-Case across all Attacks



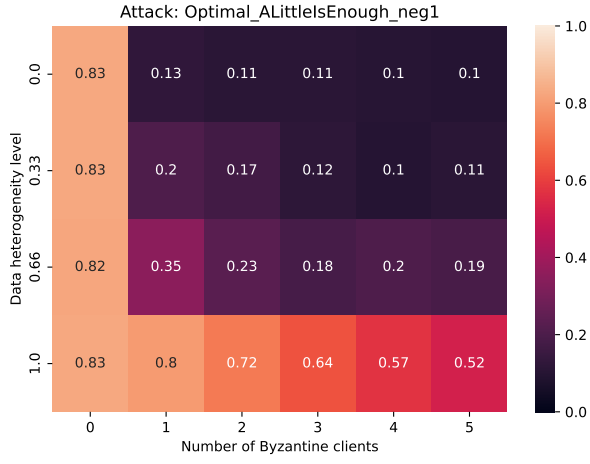
(a) CNN ReLU



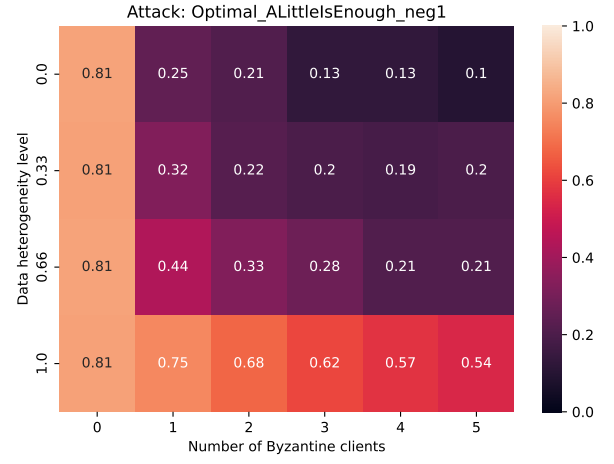
(b) CNN Tanh

Figure 5: Centered Clipping (CC) under Worst-Case across all Attacks.

3.4.2 CC under ALIE



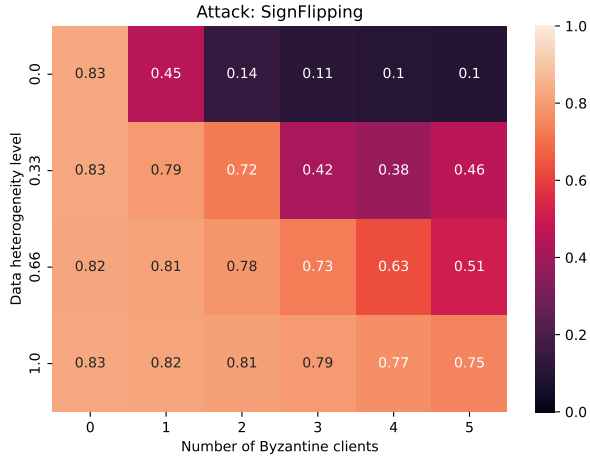
(a) CNN ReLU



(b) CNN Tanh

Figure 6: Centered Clipping (CC) under Optimal ALIE Attack.

3.4.3 CC under SF



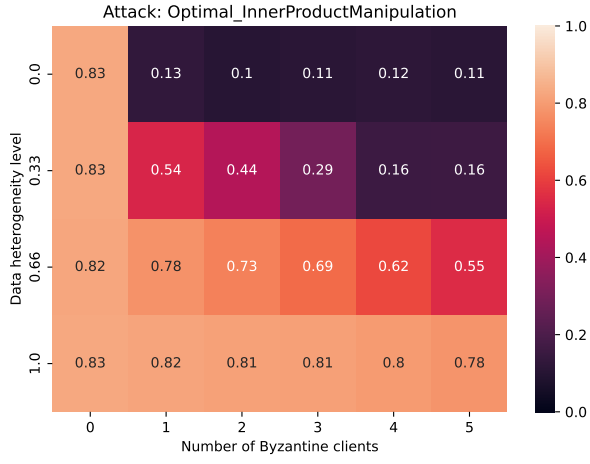
(a) CNN ReLU



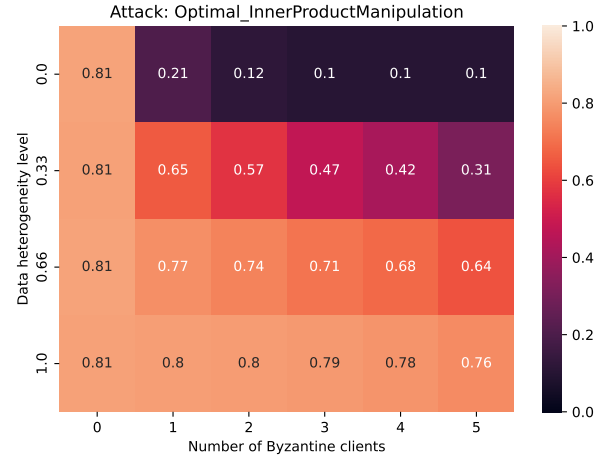
(b) CNN Tanh

Figure 7: Centered Clipping (CC) under Sign Flipping Attack.

3.4.4 CC under IPM



(a) CNN ReLU

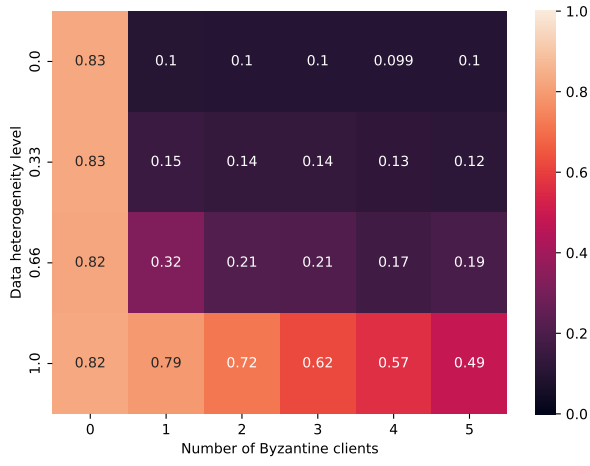


(b) CNN Tanh

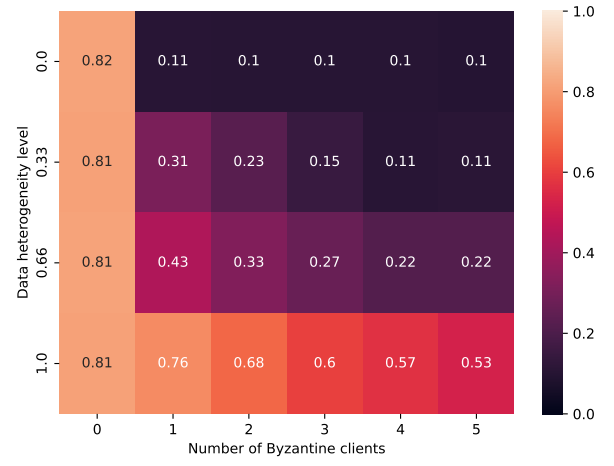
Figure 8: Centered Clipping (CC) under Optimal IPM Attack.

3.5 Geometric Median (GM)

3.5.1 GM under Worst-Case across all Attacks



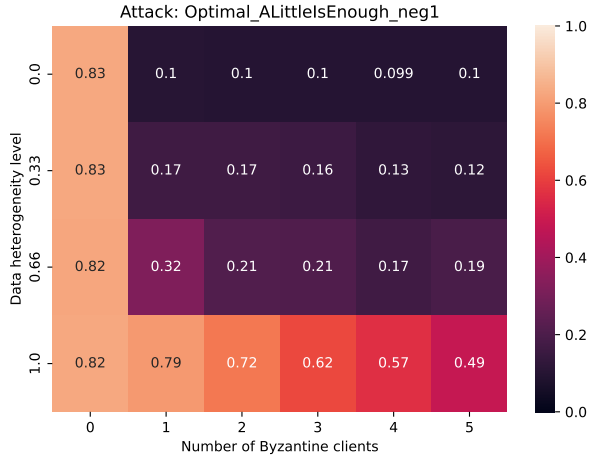
(a) CNN ReLU



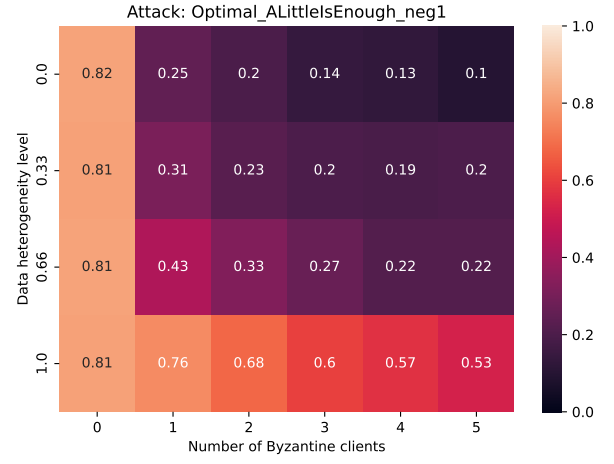
(b) CNN Tanh

Figure 9: Geometric Median (GM) under Worst-Case across all Attacks.

3.5.2 GM under ALIE



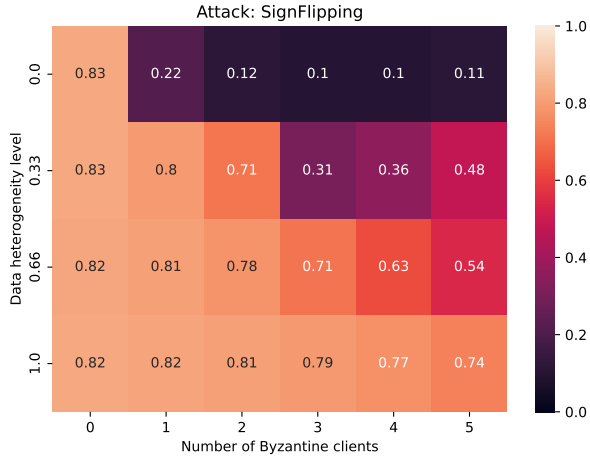
(a) CNN ReLU



(b) CNN Tanh

Figure 10: Geometric Median (GM) under Optimal ALIE Attack.

3.5.3 GM under SF



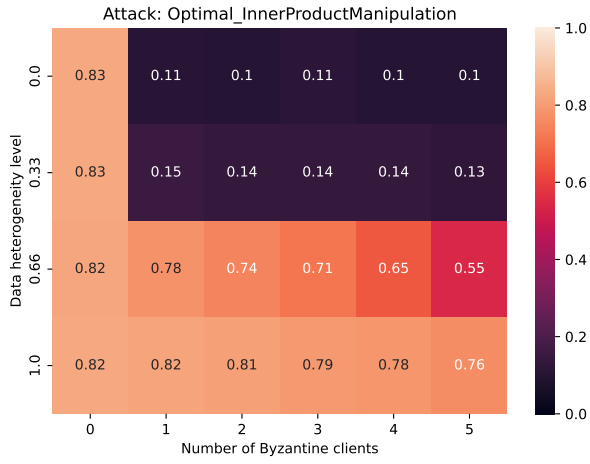
(a) CNN ReLU



(b) CNN Tanh

Figure 11: Geometric Median (GM) under Sign Flipping Attack.

3.5.4 GM under IPM



(a) CNN ReLU

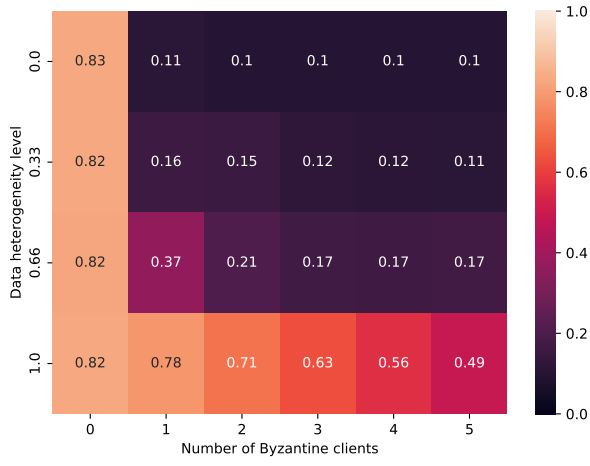


(b) CNN Tanh

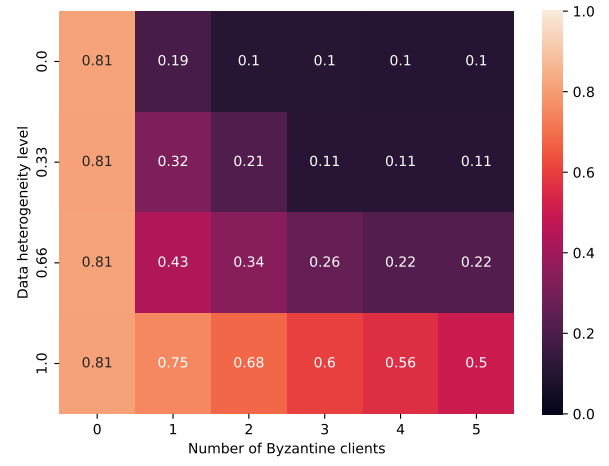
Figure 12: Geometric Median (GM) under Optimal IPM Attack.

3.6 Multi-Krum (MK)

3.6.1 MK under Worst-Case across all Attacks



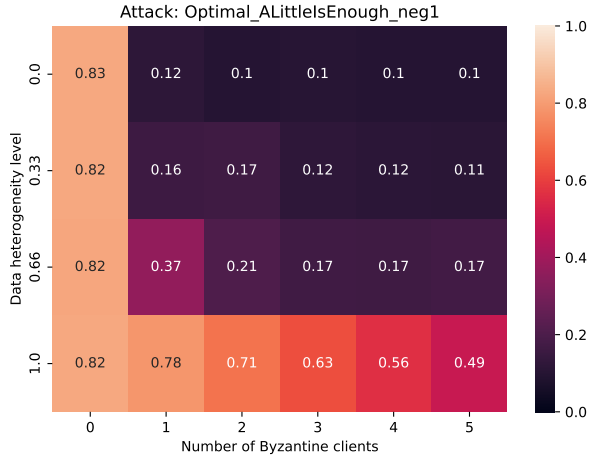
(a) CNN ReLU



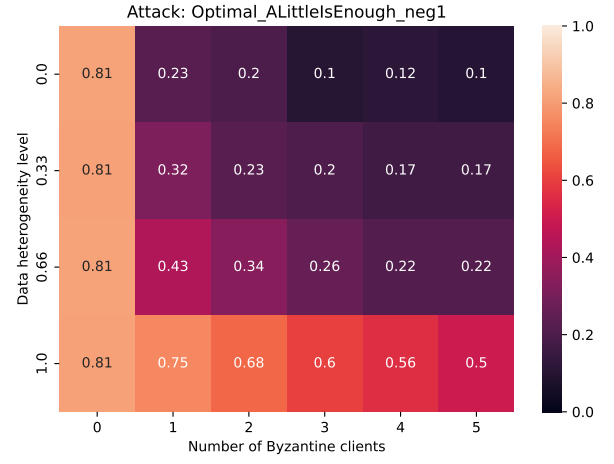
(b) CNN Tanh

Figure 13: Multi-Krum (MK) under Worst-Case across all Attacks.

3.6.2 MK under ALIE



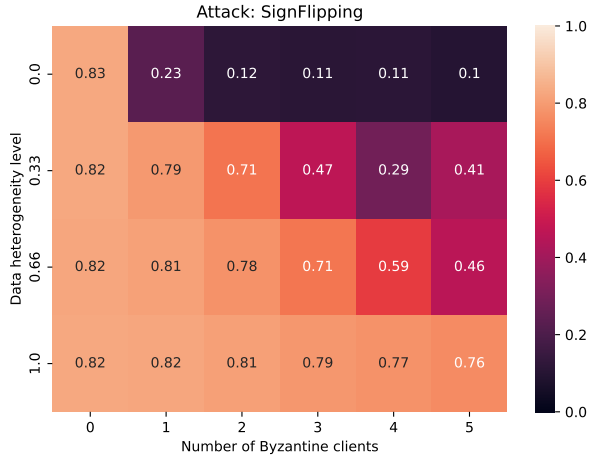
(a) CNN ReLU



(b) CNN Tanh

Figure 14: Multi-Krum (MK) under Optimal ALIE Attack.

3.6.3 MK under SF



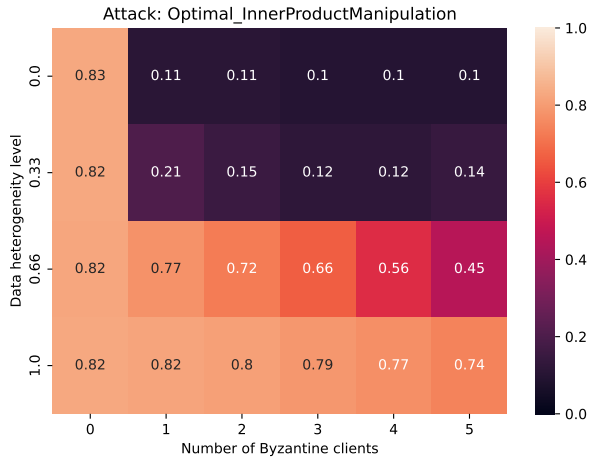
(a) CNN ReLU



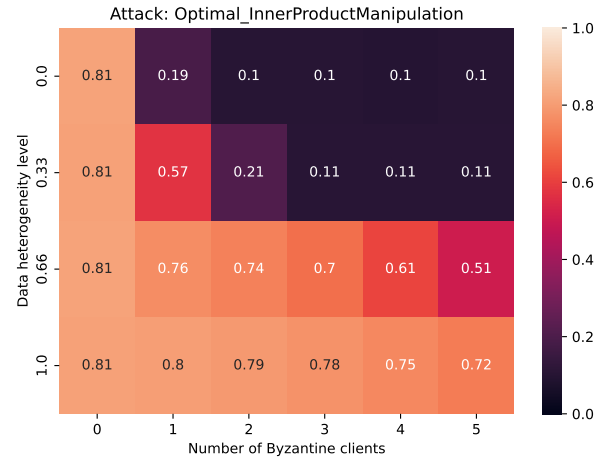
(b) CNN Tanh

Figure 15: Multi-Krum (MK) under Sign Flipping Attack.

3.6.4 MK under IPM



(a) CNN ReLU

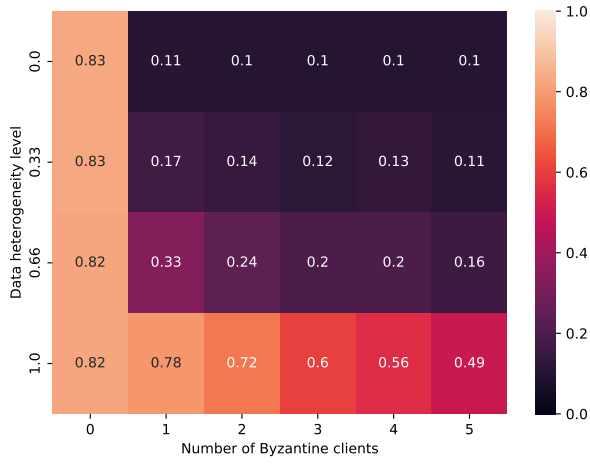


(b) CNN Tanh

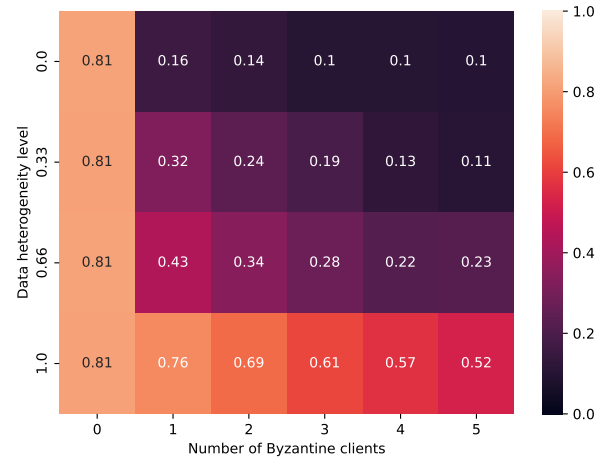
Figure 16: Multi-Krum (MK) under Optimal IPM Attack.

3.7 Trimmed Mean (TM)

3.7.1 TM under Worst-Case across all Attacks



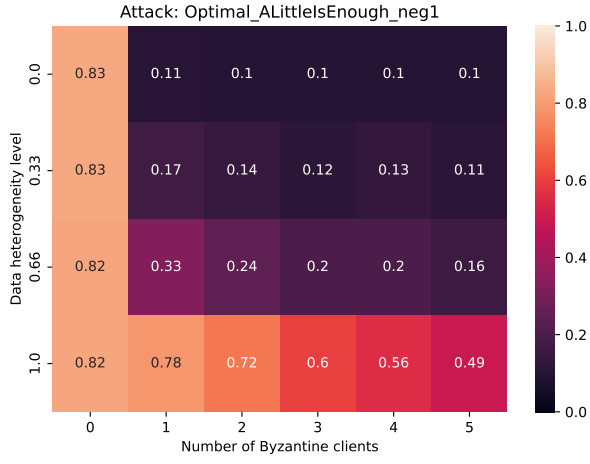
(a) CNN ReLU



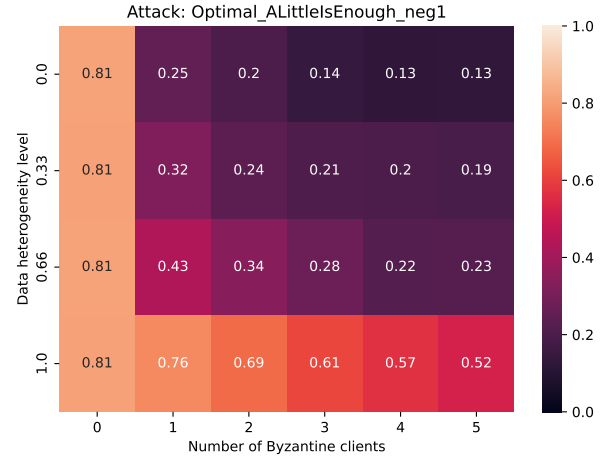
(b) CNN Tanh

Figure 17: Trimmed Mean (TM) under Worst-Case across all Attacks.

3.7.2 TM under ALIE



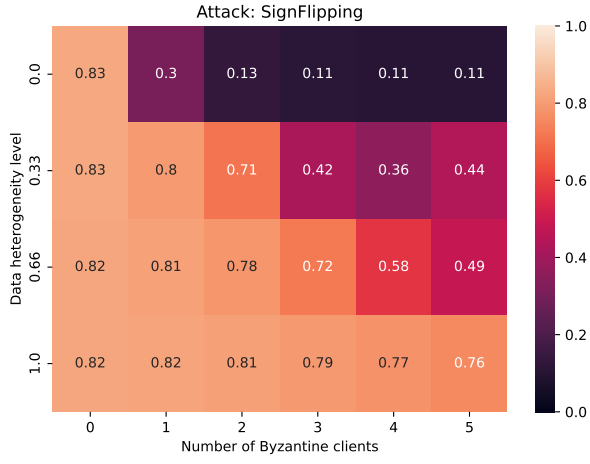
(a) CNN ReLU



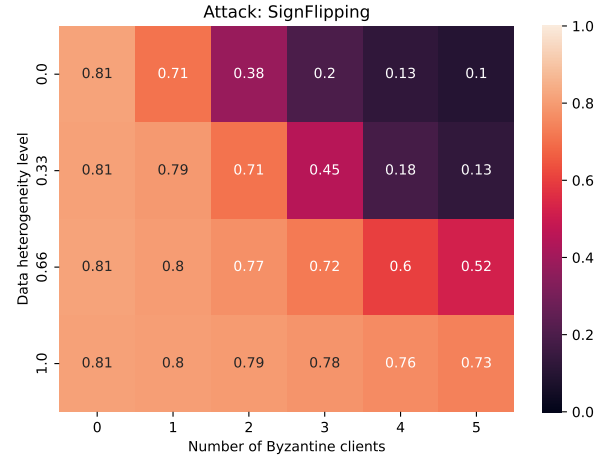
(b) CNN Tanh

Figure 18: Trimmed Mean (TM) under Optimal ALIE Attack.

3.7.3 TM under SF



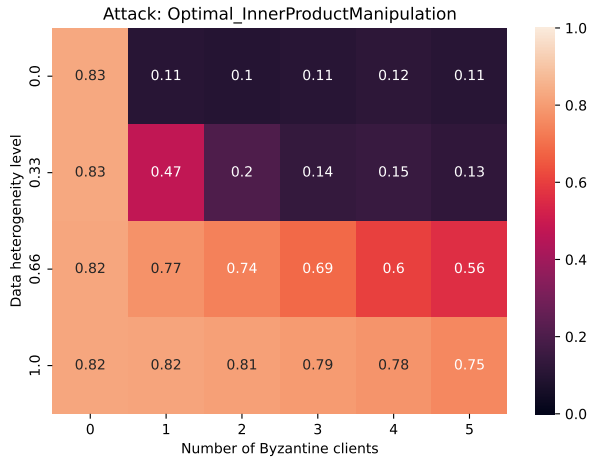
(a) CNN ReLU



(b) CNN Tanh

Figure 19: Trimmed Mean (TM) under Sign Flipping Attack.

3.7.4 TM under IPM



(a) CNN ReLU



(b) CNN Tanh

Figure 20: Trimmed Mean (TM) under Optimal IPM Attack.